

SEED STENCILS: HIGH-ORDER FINITE DIFFERENCES AND RBF-FD THROUGH MATERIAL EDGES TOO STEEP FOR THE GRID

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ABSTRACT. Between a material whose parameters vary smoothly on the scale of the grid and one with a jump lies the case of an edge that is smooth but changes over a distance δ far smaller than the grid spacing h . Standard high-order finite differences sample the coefficients at the nodes and lose their order there until $h \lesssim \delta$, and the interface-aware stencils built for a jump do not apply, since there is no jump. We build stencils on the same equispaced (1-D) or scattered (2-D) nodes whose basis functions are not monomials but *seeds*: the $t = 0$ profiles of solutions of the wave equation that are polynomial in time, continued through the edge by ordinary differential equations in the coordinate normal to it. The seeds form an extended complete Chebyshev system for every δ , so the stencil solve is nonsingular, and as $\delta \rightarrow 0$ they reduce to the translated polynomial basis of the interface-aware stencils, so one construction covers the jump, the sub-grid edge and the resolved edge. In one dimension, standard fourth-order finite differences have a knee at $h = \delta$ while the seed stencils are fourth order at every resolution with the errors of the jump case. In two dimensions, for the elastic wave equation discretised with radial basis function-generated finite differences (RBF-FD) on scattered nodes, the seeds of a straight edge reduce to triangular systems of ordinary differential equations in the normal coordinate; the seed-augmented stencils are stable at the standard hyperviscosity, cut the error through an unresolved edge by up to an order of magnitude or more on the node sets tested, and lead to the rule: seed when $\delta \leq h$. The same seeds marched along the true normal handle a curved edge at the same accuracy on those node sets.

1. INTRODUCTION

High-order finite differences for wave propagation are in good shape at both ends of the range of media. When the material parameters vary smoothly on the scale of the grid, sampling them at the nodes costs nothing and a fourth-order stencil stays fourth order. When they jump, the stencils that straddle the jump can be rebuilt from piecewise polynomials that satisfy the interface conditions, and the scheme keeps its order on a grid that takes no notice of the interface [Mar16, MF17a]. Between the two lies a case that is neither. An edge can be smooth, so that no interface condition applies, and still change over a distance δ smaller than the grid spacing

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h . Any smooth model contains such edges once it is discretised coarsely enough, and any jump becomes one under the slightest smoothing.

A standard stencil through such an edge samples the coefficients at its nodes, sees a jump, and treats it as if the coefficients were smooth in between, which is the first-order failure of the jump case. It recovers only once $h \lesssim \delta$. In the one-dimensional experiment of Section 4, fourth-order finite differences (FD4) through an edge of width $\delta = 0.0025$ converge at rates 1.3 and 2.2 while $h > \delta$, drop by a factor 200 between $n = 400$ and $n = 800$ nodes ($h = 2\delta$ to $h = \delta$), and are fourth order after that: a knee at $h = \delta$. The two existing repairs do not reach this regime. The interface-aware stencils built for a jump are the $\delta \rightarrow 0$ limit of the right construction, as Section 3 shows, but their weights are off at first order in δ/h , by 10% of the largest weight at $\delta = h/10$ and 1% at $\delta = h/100$ in the example of Section 3.2. The other route expands the coefficients in Taylor series about the edge, as the interface-aware construction can for a wave speed that varies smoothly up to an interface [Mar16, MF17a] and as polynomial quasi-Trefftz spaces do in general [IGMS23]; but a profile such as $\tanh(x/\delta)$ has poles at $\pm i\pi\delta/2$, so the expansion converges only within $\pi\delta/2$ of the edge while the stencil reaches $2h \gg \delta$.

The construction of this paper keeps everything about the scheme except the basis functions of the stencils that see the edge. In place of the monomials $1, x, x^2, \dots$ a stencil is made exact on *seeds*: the $t = 0$ profiles of solutions of the wave equation that are polynomial in time. For the operator $L_u = \rho^{-1}\partial_x K \partial_x$ that governs the particle velocity, such a profile satisfies $L_u^m g = 0$ for some m , and with constant coefficients these profiles are exactly the polynomials. Anchored at the stencil's evaluation point x_e they form a chain, $\phi_0 = 1$, $K\phi'_1 = K(x_e)$, $L_u\phi_k = k(k-1)c_e^2\phi_{k-2}$ with $\phi_k(x_e) = \phi'_k(x_e) = 0$, which is integrated through the edge as a first-order system of ordinary differential equations that never differentiates the coefficients: one short march per stencil that sees the edge. Three facts make this a method rather than a trick. The seeds form an extended complete Chebyshev system for every δ , so the per-stencil solve is nonsingular on any distinct nodes, including at $\delta = 0$ where the seeds have kinks, with the conditioning of a Vandermonde matrix (condition numbers 20 to 60 from $\delta = 10^{-5}$ to 1). The truncation error of the seed stencil on a solution is $O(h^4 \partial_x \partial_t^4 u)$, bounded independently of δ , where FD4's $O(h^4 u^{(5)})$ carries $u^{(5)} \sim \delta^{-4}$ inside the edge. And the chain read in the weak sense across a jump is the translated Taylor basis of the interface-aware stencils, so one construction covers the jump (algebra), the sub-grid edge (an ODE march) and the resolved edge, where the seeds and the monomials give different weights, first order in h/δ apart, and both are fourth order on solutions.

The same idea carries to two dimensions, to the elastic wave equation on scattered nodes discretised with radial basis function-generated finite differences (RBF-FD). For a straight edge the material depends on the normal coordinate only, and an ansatz polynomial in the tangential coordinate turns

the chain into triangular systems of one-dimensional ODEs in the normal coordinate: all 30 seeds of one stencil, the monomials to degree four in each velocity component, march as one linear system of 600 states anchored at the evaluation node. The seeds replace the polynomial augmentation of the interface-aware RBF-FD stencils; the Gaussian part of the stencil and the coupled saddle-point solve stay as they are. For a curved edge the same seeds are marched along the true normal through the stencil's foot point, an approximation of zeroth order in the curvature, which is the one the jump stencils already make.

Contributions.

- (1) In one dimension: the seed chain, its nested kernels, its well-posedness as an extended complete Chebyshev system, and the jump construction as its $\delta \rightarrow 0$ limit (Section 3). On the same equispaced grid the seed stencils are fourth order at every resolution, with errors matching the jump case to three digits, while FD4 has its knee at $h = \delta$ (Section 4).
- (2) In two dimensions: the elastic seeds of a straight edge as triangular systems of ODEs in the normal coordinate, marched as one system per stencil; their assembly into seed rows for the elastic operator and hyperviscosity rows that annihilate the seeds; and the finding that the seeded operator is stable at the standard hyperviscosity at every edge width (Sections 5 and 6).
- (3) A rule: *seed when $\delta \leq h$* . Through an edge the nodes never resolve the seeds are fourth order at every resolution tested, $20\times$ below the standard scheme in the velocity and $67\times$ in the spurious tangential velocity at 19,600 nodes, at the seed operator's own resolution floor; the crossover is at $h \approx \delta$; and through a resolved edge the seed stencils are 1.4 to $2\times$ worse than the standard ones, so a resolved edge should not be seeded.
- (4) At oblique incidence the tangential dependence of the seeds is what acts: the full seeds beat the standard scheme by 1.4 to $1.9\times$ and the normal-only ablation is worse than doing nothing through a sharp edge. Every scheme converges at about order 2.5 there, the jump-aware stencils included, because the mode-converted shear waves sit at a pre-asymptotic floor on these node sets; the result is reported as it is.
- (5) Through a curved edge the jump stencils sit at the resolution floor at every node count, and the seeds marched along the true normal equal the flat seeds at their own floor, 2.2 to $7\times$ below the standard scheme at $\delta = 0.005$ and 3 to $12\times$ at $\delta = 0.0025$ on node sets up to 19,600 nodes; the zeroth-order geometry first shows at the finest of them.

- (6) An open implementation in Python with a test for every numerical routine, in which every figure and table of this paper is regenerated by a committed script from a results cache.

1.1. Relation to prior work. Three lines of prior work meet here, and the construction adds nothing to any of them taken alone. Stencils and elements built from local solutions of the operator are the fitted operators of the 1950s and 60s [AS55, Il'69, SG69, RST08] and the exact schemes that followed [Mic93, VSKO20]; the constant-flux seed in particular is the harmonic mean of Tikhonov and Samarskii [TS62, Sam01], the special elements of Babuška and Osborn [BO83, BCO94], the multiscale elements of Hou and Wu [HW97] and the harmonic coordinates of Owhadi and Zhang [OZ07], whose extension to the acoustic wave equation [OZ08] is the closest two-dimensional relative of what follows. Basis functions that are polynomial-in-time solutions of the wave equation are the Trefftz spaces of space–time discontinuous Galerkin methods [KMPS16, BGL17, MP18]; their quasi-Trefftz extension to variable coefficients [IGMS23, IGMP25, IG25] builds them by Taylor expansion at a point, the route Section 3.4 shows cannot reach an edge thinner than the stencil. For a jump, the interface-aware stencils this work generalises [Mar16, MFSC15, MF17a] stand beside the immersed interface [LL94, ZL97, SMP⁺23], explicit simplified interface [PL01, LP04] and summation-by-parts [MN06, DV14, GW25] treatments, all of which take the jump as given, where standard schemes are first order [Bro84, SV09, VLTR14]. The standing alternative in seismic and electromagnetic finite differences changes the medium rather than the stencil: coefficients averaged over a cell [Bac62, MDEN92, MKV⁺02, KMCK17, JZ24, FRR⁺06], thin structure homogenised [CGM10a, CGM10b], or the jump smoothed over a few cells [TE03, TE06, KWR22]. Those methods choose the smoothing width from the grid and accept a second-order error component at the interface [TE06]; here the width is a property of the medium, smaller than the grid, and the aim is the scheme's full order through it. We found no prior instance of stencils on an unchanged equispaced or scattered node set whose basis is continued through a sub-grid smooth edge by ordinary differential equations, nor of the observation that the construction reduces to the jump stencils as $\delta \rightarrow 0$; the search is logged in the repository's `LITERATURE.md`. We did not compare against the smoothing and averaging methods, and claim nothing relative to them.

Outline. Section 2 fixes the equations, the discretisations and the interface-aware stencils of the jump case that the seeds extend. Section 3 constructs the seeds in one dimension and proves what can be proved; Section 4 is the one-dimensional experiment. Sections 5 and 6 do the same in two dimensions, flat edges first, then oblique incidence and curved edges. Section 7 records the limitations and what we would do next.

2. SETTING

2.1. One dimension. The one-dimensional problem is the two-way wave equation in first-order form on the periodic interval $[-1, 1)$, with u the particle velocity and f the stress,

$$(1) \quad \rho u_t = f_x, \quad f_t = K u_x, \quad K = \rho c^2,$$

where $c(x) > 0$ is the wave speed and $\rho(x) > 0$ the density. Eliminating one field gives a second-order operator for each,

$$(2) \quad u_{tt} = L_u u, \quad L_u = \frac{1}{\rho} \partial_x K \partial_x; \quad f_{tt} = L_f f, \quad L_f = K \partial_x \frac{1}{\rho} \partial_x.$$

The medium is a background material $(c, \rho) = (1, 1)$ with one layer $[x_1, x_2) = [0, \frac{1}{2})$ of $(c, \rho) = (2, 1)$, so the contrast in K is 4 and the impedance ρc doubles. With edge width $\delta > 0$ the two edges of the layer are tanh transitions,

$$(3) \quad s(x) = \frac{1}{2} \sum_{m \in \mathbb{Z}} \left[\tanh \frac{x - x_1 + 2m}{\delta} - \tanh \frac{x - x_2 + 2m}{\delta} \right],$$

$$c = c_{\text{bg}} + (c_{\text{layer}} - c_{\text{bg}}) s, \quad \rho \text{ likewise,}$$

summed over the periodic images so that the profile is smooth and periodic to rounding; $\delta = 0$ is the jump. For $\delta \ll h$ a stencil sees a jump between two of its nodes; for $\delta \gg h$ it sees a smooth function; the case in between is the subject of the paper.

The grid has n cell-centred nodes, $x_i = -1 + (i + \frac{1}{2})h$ with $h = 2/n$, and the semi-discrete system is

$$(4) \quad \rho_i \dot{u}_i = (D_f f)_i, \quad \dot{f}_i = K_i (D_u u)_i,$$

with $\rho_i = \rho(x_i)$, $K_i = K(x_i)$ and D_u, D_f two differentiation matrices with the sparsity of the centred five-point stencil. In the *naïve* scheme both are the standard fourth-order matrix, Fornberg's weights on the monomials [For88a]: the coefficients are sampled at the nodes and nothing else is done about the edge. In the *seed* scheme the rows whose stencils see a varying material are rebuilt as in Section 3, separately for u and for f because the two fields carry different continuity conditions; every other row is the standard one. Time stepping is the classical fourth-order Runge–Kutta method (RK4) with $\Delta t = 0.4 h / c_{\text{max}}$. Errors are relative ℓ^2 errors in the stress f at $t = 1$ against a reference solution (Section 4).

2.2. Two dimensions. The two-dimensional problem is the isotropic elastic wave equation in first-order form on the doubly periodic unit square, with velocities (u, v) and stresses $(\sigma_{xx}, \sigma_{xy}, \sigma_{yy})$,

$$(5) \quad \begin{aligned} \rho u_t &= \partial_x \sigma_{xx} + \partial_y \sigma_{xy}, & \partial_t \sigma_{xx} &= (\lambda + 2\mu) u_x + \lambda v_y, \\ \rho v_t &= \partial_x \sigma_{xy} + \partial_y \sigma_{yy}, & \partial_t \sigma_{xy} &= \mu (u_y + v_x), \\ & & \partial_t \sigma_{yy} &= \lambda u_x + (\lambda + 2\mu) v_y, \end{aligned}$$

with Lamé parameters $\lambda, \mu > 0$ and density $\rho > 0$, P- and S-wave speeds $c_p = \sqrt{(\lambda + 2\mu)/\rho}$ and $c_s = \sqrt{\mu/\rho}$, and $K = \lambda + 2\mu$ in the role of ρc^2 . (The implementation and the dissertation name the stresses f, g, h ; we write σ so that h can stay the node spacing.) The medium is a background $(\lambda, \mu, \rho) = (1, 1, 1)$ with a band $(4, 4, 2)$ between the two interfaces $y = y_0 + A \sin 2\pi x$, $y_0 = \frac{1}{4}$ and $\frac{1}{2}$, so c_p rises from $\sqrt{3}$ to $\sqrt{6}$ and c_s from 1 to $\sqrt{2}$ across each interface. The flat case is $A = 0$ and the curved case $A = 0.02$, the two geometries of the dissertation's elastic tests [Mar16]. With edge width $\delta > 0$ each interface becomes a tanh transition in the signed normal distance d to it, with the band's weight s summed over the periodic images in y ,

$$(6) \quad \begin{aligned} s &= \frac{1}{2} [1 + \tanh(d/\delta)], \\ (\lambda, \mu, \rho) &= (\lambda, \mu, \rho)_{\text{bg}} + [(\lambda, \mu, \rho)_{\text{band}} - (\lambda, \mu, \rho)_{\text{bg}}] s, \end{aligned}$$

so that λ, μ and ρ , and hence K , are linear in the weight while c_p and c_s are not. For a flat interface d is the vertical offset; for a curved one it is the true signed distance, through the foot point on the curve. Two edges a gap g apart reach only $\tanh(g/2\delta)$ of the contrast between them, so the band's width of $\frac{1}{4}$ bounds the edge widths we can pose; nothing here approaches that bound.

Node sets. A node set of n nodes is built as in [MF17a]: six fixed rows of nodes hug each interface, three on each side at normal distances $(\frac{1}{2} + j\frac{\sqrt{3}}{2})h$, $j = 0, 1, 2$, with alternate rows shifted by half a spacing so that the rows straddle the interface in a hexagonal pattern, and every other node is placed by a simulated electrostatic repulsion from a random start. The straddling rows were found empirically to be what keeps the interface-aware stencils stable, and we keep them for the seed stencils. Throughout, $h = n^{-1/2}$ denotes the mean spacing, and convergence rates are measured in it; the node counts are $n = 2500, 4900, 10,000$ and $19,600$, so h runs from 0.02 to 0.0071.

RBF-FD. For a stencil of s nodes $x_1, \dots, x_s$ about an evaluation node x_c , the weights w of a linear operator $\mathcal{L}$ solve the saddle-point system

$$(7) \quad \begin{bmatrix} A & P \\ P^\top & 0 \end{bmatrix} \begin{bmatrix} w \\ w_* \end{bmatrix} = \begin{bmatrix} \mathcal{L} \varphi(|x - x_i|)|_{x_c} \\ \mathcal{L} p_m(x)|_{x_c} \end{bmatrix},$$

with $A_{ij} = \varphi(|x_i - x_j|)$ for the Gaussian $\varphi(r) = e^{-(\varepsilon r)^2}$, $P_{im} = p_m(x_i)$ over the bivariate monomials p_m up to a chosen degree, evaluated in coordinates scaled by the stencil radius, and w_* discarded. The constraint rows make the weights exact on the polynomials, which take over from the Gaussians under refinement, and the shape parameter is scaled to the local spacing, $\varepsilon = 0.4/d$ with d the distance from x_c to its third-nearest neighbour.¹ Away from the

¹The published description [MF17a] says the nearest neighbour; the original MATLAB and the present implementation use the third-nearest. The choice fixes a constant in εh and nothing else.

interfaces every stencil has 30 nodes and is exact on the polynomials of degree ≤ 4 ; stencils across a jump have 19 nodes and degree ≤ 3 , with the piecewise basis of Section 2.3. Two matrices D_x and D_y result, and the $5n \times 5n$ block operator of (5) is assembled from them with the material coefficients evaluated at each row's node. A small hyperviscosity $\gamma\Delta^3$ is added to every field to move the spurious eigenvalues of the RBF-FD operator into the left half-plane [FL11]: the Δ^3 stencils are built on the same nodes and the same basis as the derivative stencils, and $\gamma = \gamma_0 (h/h_0)^5$ with $\gamma_0 = 2.4 \times 10^{-11}$ at $h_0 = 0.02$, which keeps $\gamma\Delta^3$ proportional to $1/h$ like the wave operator so that one CFL number serves every n . Time stepping is RK4 with $\Delta t = 0.5 h/c_{p,\max}$, capped where necessary by the hyperviscosity spectrum.

Pulses and errors. The normal-incidence runs send a plane P-wave pulse, Gaussian in y with sharpness 15, from $y = 0.875$ in the background towards $-y$ through both interfaces; the true solution then has $u \equiv 0$, and the largest $|u|$ a scheme produces is a clean measure of what its stencils do at the edges. The oblique runs send a plane-wave train at 26.6° to the normal (direction $(1, 2)$), which mode-converts at each interface. Errors are relative ℓ^2 errors in v at $t = 1$ against the references described with each experiment in Section 6, together with the spurious u at normal incidence and the relative error in u otherwise.

2.3. Interface-aware stencils for a jump. The seeds extend the interface-aware stencils of [Mar16, MF17a], which we summarise in the form the later sections use; the derivations are in those references.

One dimension. Write (1) as $\partial_t(u, f)^\top = \mathcal{D}(u, f)^\top$ with the block operator

$$\mathcal{D} = \begin{bmatrix} 0 & \rho^{-1}\partial_x \\ K\partial_x & 0 \end{bmatrix}.$$

Across a material jump, u and f are continuous for all time, hence so are all their time derivatives, and $\partial_t^k(u, f)^\top = \mathcal{D}^k(u, f)^\top$ converts each of those conditions into a condition on spatial derivatives, with the material of the side it is evaluated on: $\mathcal{D}_L^k(u_L, f_L)^\top = \mathcal{D}_R^k(u_R, f_R)^\top$ at the jump for $k = 0, 1, 2, \dots$. Expanding u and f on each side as polynomials of degree four about the jump and imposing these conditions for $k \leq 4$ gives a square *continuity matrix* per side, $C_L a_L = C_R a_R$ on the coefficient vectors. Keeping the monomials on one side and translating them across with $C_R^{-1}C_L$ yields the *translated basis*: piecewise polynomials, one per monomial, that satisfy every continuity condition. A stencil whose nodes straddle both edges of a layer thinner than the stencil chains two translations. The stencil weights are then the ones that differentiate every basis function exactly at the evaluation point, a small dense solve per stencil, and the f basis is built by the same construction rather than from the u basis through the operator, which would lose an order. This is the scheme that Section 4 calls the jump case; the even continuity conditions relate a field to itself and the odd ones route through the other field, a pattern the seeds inherit.

Two dimensions. A stencil near an interface works in a local frame: origin at the nearest interface point, x' along the tangent and y' along the normal. The isotropic system (5) keeps its form under rotation, so in that frame the rotated fields obey the same equations and the interface is, locally, $y' = 0$. The velocities (u', v') and the tractions $(\sigma'_{x'y'}, \sigma'_{y'y'})$ are continuous across it for all time; the remaining stress $\sigma'_{x'x'}$ may jump. Expand (u', v') on each side in the monomials of (x', y') up to degree four, convert the time derivatives of the continuous quantities through powers of the block operator (even powers relate velocities to velocities, odd powers route through the stresses), and match the coefficients of the powers of x' at each order: a square continuity matrix per side, a translated velocity basis, truncated to degree three. The stress basis is generated from it by applying the stress rows of the operator on each side and dropping the three columns that come from rigid motions, 27 functions in all. The RBF-FD solve (7) then couples the fields: the stress rates need weights on u' and v' data jointly, the velocity rates weights on the three stresses jointly, with the Gaussian block repeated on the diagonal and the piecewise basis in place of P . The weights are rotated back to the laboratory frame, and the hyperviscosity stencils across the interface use the same piecewise basis, which the original work found necessary for accuracy. The interface is treated as locally flat in the frame; carrying curvature into the continuity conditions by expanding the interface in x' is possible [MF17a] but was not used by the original code and is not used here. This is the approximation we call zeroth order in the curvature, and the curved seeds of Section 5 make the same one.

2.4. Scope of the implementation. The experiments run on a Python reimplementations of the 2016–2017 methods (NumPy and SciPy, with a test for every numerical routine) that reproduces the constructions above and restructures the code. Of the original two-dimensional code it leaves out three things: the *triple stencils* that see both interfaces of a band thinner than one stencil (the band here is $\frac{1}{4}$ wide and none is needed), the variable Lamé parameters inside the band of the published test case (the band here is a constant material), and the finite-difference far field of the FD/RBF-FD hybrid (every stencil here is RBF-FD). Point sources and the Marmousi model are out of scope. In one dimension the implementation takes piecewise-constant materials, with the smooth edges of (3), and not the dissertation’s smoothly varying background, because the exact reference of the jump case, a ray sum, needs constant pieces; the published treatment of a wave speed that varies smoothly up to an interface, a degree-four Taylor expansion of c about the interface point [Mar16, MF17a], is the Taylor route of Section 3.4 and is not ported. Everything is fourth order (FD4 in one dimension, degree four away from and degree three across the edges in two) at one material contrast; Section 7 returns to both restrictions.

3. SEEDS IN ONE DIMENSION

Fix a stencil of the grid of Section 2.1: nodes $x_1 < \dots < x_p$, $p = 5$ for FD4, and an evaluation point x_e , one of the nodes. The standard weights are the ones that differentiate the monomials $(x - x_e)^k$, $k < p$, exactly at x_e ; the interface-aware weights of Section 2.3 do the same for the translated basis. This section builds the basis for a smooth edge and shows that the two are its limits.

3.1. Seeds as the profiles of solutions polynomial in time. Consider the velocity field with its operator L_u of (2) and look for solutions of $u_{tt} = L_u u$ that are polynomial in time, $u(x, t) = \sum_{j=0}^J t^j g_j(x)$. Matching powers of t gives $(j+2)(j+1)g_{j+2} = L_u g_j$, so the two top coefficients lie in $\ker L_u$ and the $t = 0$ profile g_0 satisfies $L_u^m g_0 = 0$ for some m . With constant coefficients $L_u = c^2 \partial_x^2$ and $\ker \partial_x^{2m}$ is exactly the polynomials of degree $< 2m$: the profiles of time-polynomial solutions are what plays the role of polynomials for the operator. We fix a basis of them by anchoring at the evaluation point.

Definition 3.1 (Seeds). Let x_e be the evaluation point, $c_e = c(x_e)$ and $K_e = K(x_e)$. The *seeds* of the u field at x_e are the functions $\phi_0, \phi_1, \phi_2, \dots$ with

$$(8) \quad \begin{aligned} \phi_0 &= 1, & K\phi_1' &= K_e, & \phi_1(x_e) &= 0, \\ L_u \phi_k &= k(k-1)c_e^2 \phi_{k-2}, & \phi_k(x_e) &= \phi_k'(x_e) = 0 & (k \geq 2). \end{aligned}$$

The seeds of the f field are defined by the same chain with L_f in place of L_u and $\phi_1'/\rho = 1/\rho_e$ in place of the flux condition.

Each seed is the profile of one time-polynomial solution, and the normalisation is chosen so that $\phi_k = (x - x_e)^k$ when the coefficients are constant: $L_u(x - x_e)^k = c^2 k(k-1)(x - x_e)^{k-2}$, and the anchor conditions fix the two integration constants. Read one at a time, the chain says what each monomial becomes through the edge. The constant solution $u \equiv C$ gives ϕ_0 . A solution linear in time, $u = g_0 + tg_1$ with $u_t = g_1 = C$, forces $L_u g_0 = 0$, that is $Kg_0' = \text{const}$: the x -like seed ϕ_1 is the profile of *constant flux*, $\phi_1(x) = K_e \int_{x_e}^x d\xi/K(\xi)$, and through the other field this says that $f_t = Ku_x$ is constant in x . It is the flux that stays smooth through the edge, not the slope. A solution with $u_{tt} = C$ forces $L_u g_0 = C$, and $C = 2c_e^2$ with the anchor conditions gives the x^2 -like seed; here $C \neq 0$ matters, since $C = 0$ would only return $\ker L_u$. The cubic condition $u_{ttt} = C$ fixes the profile of u_t , not of u , so ϕ_3 is the profile whose second time derivative is x -like, $L_u \phi_3 = 6c_e^2 \phi_1$; through the other field, $f_{ttt} = K(u_{tt})_x = 6c_e^2 K_e$ is constant. The even seeds are thus the conditions $\partial_t^k u = \text{const}$ and the odd seeds the conditions $\partial_t^k f = \text{const}$, the same alternation between the fields as in the continuity conditions of the jump case, where the even conditions relate a field to itself and the odd ones route through the other field.

First-order form. The chain is integrated as a first-order system that never differentiates the coefficients. For the u seeds the state is (ϕ_k, ψ_k) with $\psi_k = K\phi'_k$ the flux,

$$(9) \quad \phi'_k = \psi_k/K, \quad \psi'_k = \rho k(k-1) c_e^2 \phi_{k-2},$$

with initial data $\phi_0(x_e) = 1, \psi_1(x_e) = K_e$ and zero otherwise; for the f seeds the state is (ϕ_k, ψ_k) with $\psi_k = \phi'_k/\rho, \phi'_k = \rho\psi_k, \psi'_k = k(k-1) c_e^2 \phi_{k-2}/K$, and $\psi_1(x_e) = 1/\rho_e$. All p seeds of a field march together, from x_e outward to each side of the stencil, since the right-hand side of ϕ_k only involves ϕ_{k-2} . In the stencil coordinate $\xi = (x - x_e)/h_s$, with h_s the stencil half-width, the chain reads the same with ∂_x replaced by ∂_ξ , and then $\phi_k \approx \xi^k$: the interpolation matrix $A_{kj} = \phi_k(\xi_j)$ is a small perturbed Vandermonde matrix, and its condition number stays between 20 and 60 for δ from 10^{-5} to 1 in the setting of Section 2.1. Only ϕ_1 has a non-zero derivative at the anchor, so the weights solve $A\tilde{w} = e_1$ and $w = \tilde{w}/h_s$. Figure 1 shows the u seeds of one stencil for three edge widths.

3.2. Nested kernels and the jump as the limit. Only the span of the seeds enters the weights, and the spans are canonical. Since L_u sends each seed two steps down the chain, $\text{span}\{\phi_0, \dots, \phi_k\}$ is the $(k+1)$ -th space of

$$(10) \quad \{1\} \subset \ker L_u \subset \{L_u g = \text{const}\} \subset \ker L_u^2 \subset \{L_u^2 g = \text{const}\} \subset \dots,$$

where $\{L_u g = \text{const}\}$ is short for the space of g with $L_u g$ constant, so that a different anchor or normalisation changes the basis and not the space, and the weights not at all.

For a jump, read L_u in the weak sense: ϕ and the flux $K\phi'$ continuous across it. Then $\ker L_u$ is spanned by 1 and a piecewise-linear function whose slopes are in the ratio K_L/K_R , which is the translated x of Section 2.3, the function that keeps $\rho c^2 u_x$ continuous; $L_u \phi_2 = 2c_e^2$ gives $\phi_2'' = 2c_e^2/c^2$ on each side, which is the continuity of u_{tt} ; and so on up the chain, each condition $\partial_t^k u = \text{const}$ or $\partial_t^k f = \text{const}$ being one of the continuity conditions the translated basis was built from. The chain for a jump is therefore the translated Taylor basis, and the interface-aware stencils are the seed stencils at $\delta = 0$: one construction with two implementations, algebra for a jump and an ODE march for a smooth edge. The limit is approached at first order in δ . For a stencil of half-width 0.02 straddling an edge of the default contrast, the seed weights differ from the interface-aware weights by 22%, 10%, 0.98% and 0.098% of the largest weight at $\delta = 10^{-2}, 10^{-3}, 10^{-4}$ and 10^{-5} , for both fields, and equally for a stencil that straddles both edges of a layer thinner than itself, the double-cross of the jump case, which the march handles by passing through both edges.

3.3. Well-posedness. The per-stencil solve $A\tilde{w} = e_1$ is nonsingular for every edge width, including the jump, and not only for the node sets we happened to use.

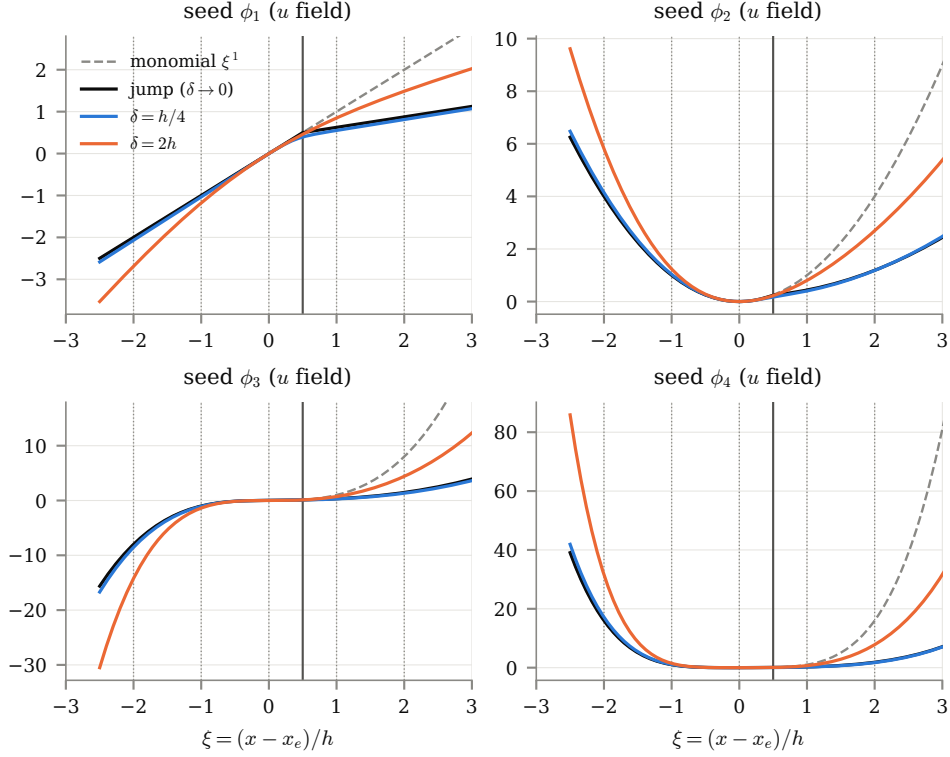


FIGURE 1. The u seeds $\phi_1, \dots, \phi_4$ of one five-point stencil whose evaluation point lies $h/2$ to the left of the lower edge of the layer of Section 2.1 (K from 1 to 4), for a jump, for $\delta = h/4$ and for $\delta = 2h$, against the monomials. The jump seeds are the translated Taylor basis of the interface-aware stencils: a slope ratio $K_L/K_R = 1/4$ for ϕ_1 , a curvature ratio $c_L^2/c_R^2 = 1/4$ for ϕ_2 . The seeds for $\delta = h/4$ are indistinguishable from them, the seeds for $\delta = 2h$ bend over the whole stencil, and all of them leave the monomials at the edge.

Proposition 3.2. *Let K and ρ be positive and continuous on an interval I containing x_e (or positive and piecewise continuous, with L_u read in the weak sense). Then for every k the seeds $\phi_0, \dots, \phi_k$ of Definition 3.1 form an extended complete Chebyshev system on I : interpolation in their span at any $k+1$ distinct points of I is uniquely solvable. The same holds for the f seeds.*

Proof. Each space in (10) is the kernel of an operator in Pólya form, a product of ∂_x and multiplications by positive functions:

$$\begin{aligned}
 (11) \quad & \rho L_u^m = \partial_x K \partial_x \frac{1}{\rho} \partial_x K \partial_x \cdots \partial_x K \partial_x \quad (2m \text{ derivatives}), \\
 & \partial_x L_u^m = \partial_x \frac{1}{\rho} \partial_x K \partial_x \cdots \partial_x K \partial_x \quad (2m+1 \text{ derivatives}),
 \end{aligned}$$

with $\ker L_u^m = \ker(\rho L_u^m)$ the even spaces and $\{g : L_u^m g = \text{const}\} = \ker(\partial_x L_u^m)$ the odd ones; for L_f the roles of K and $1/\rho$ are swapped. The kernel of such an operator with positive weights $w_0, w_1, \dots$ has the nested basis $w_0, w_0 \int w_1, w_0 \int w_1 \int w_2, \dots$, and is an extended complete Chebyshev system on any interval where the weights are positive [P22, KS66, Cop71]; in particular it has the Haar property, so interpolation at distinct points in it is uniquely solvable. It remains to see that the seeds span these kernels. Their number matches the order of the operator, and they are independent because their jets at x_e are triangular: by induction on the chain, $(K\phi'_k)' = \rho k(k-1)c_e^2 \phi_{k-2}$ vanishes to order $k-2$ at x_e , so with $\phi_k(x_e) = \phi'_k(x_e) = 0$ the seed ϕ_k vanishes to order exactly k there. For the weak reading, the integrals in the nested basis are unaffected by finitely many jumps of the weights, and the argument is the same. $\square$

Interpolation in an extended complete Chebyshev system also has a Newton form with a remainder in which the operator that annihilates the space takes the place of the $(k+1)$ -th derivative (generalised divided differences, [M73]). For the five-point stencil the annihilator of $\text{span}\{\phi_0, \dots, \phi_4\}$ is $\partial_x L_u^2$, so the truncation error of the seed stencil on a function g is $O(h^4 \partial_x L_u^2 g)$. On a *solution* of the wave equation, $L_u u = u_{tt}$ and hence $\partial_x L_u^2 u = \partial_x \partial_t^4 u$, the x -derivative of another solution, which is bounded independently of δ : u_x changes by $O(1)$ across the edge whatever its width, and it is the flux Ku_x that is smooth. Standard FD4 has the error $O(h^4 u^{(5)})$ with $u^{(5)} \sim \delta^{-4}$ inside the edge. This is the prediction that Section 4 tests: standard FD4 converges at fourth order only once $h \lesssim \delta$ and looks low-order above that, with a knee at $h \approx \delta$; the seed stencils converge at fourth order with a constant that does not depend on δ .

Remark 3.3. The seeds are close relatives of two classical objects. Splines whose pieces lie in $\ker L$ for a differential operator L are the L-splines of [SV67], and the harmonic-mean coefficient $h/\int dx/K$ of the conservative difference schemes of [TS62] is the two-point version of ϕ_1 . Section 1.1 places the construction among them.

3.4. Why the Taylor route does not reach a sub-grid edge. The interface-aware construction of Section 2.3 can take smoothly varying coefficients: the continuity conditions are built from multiplication matrices formed from the Taylor coefficients of $1/\rho$ and K about the interface [Mar16, MF17a], and polynomial quasi-Trefftz spaces [IGMS23] do the same in general, with basis functions that satisfy the variable-coefficient equation to Taylor order at a point. For a sub-grid edge this route cannot work, and the reason is not the algebra. The profile $\tanh(x/\delta)$ has poles at $\pm i\pi\delta/2$, so its Taylor series about the edge converges only within a distance $\pi\delta/2$ of it, while the stencil reaches $2h \gg \delta$; no polynomial degree that the stencil could carry represents the coefficients across it. The ODE march is what a sub-grid edge needs. For a resolved edge, $\delta \gg h$, the seed weights still

differ from Fornberg’s at first order in h/δ , by 17% of the largest weight at $\delta = 8h$ for the default contrast, because the seeds are exact for a different five-dimensional space; both stencils are fourth order on solutions there, and Section 4 finds them indistinguishable end to end.

3.5. Stencil assembly and cost. The seed scheme rebuilds every row of D_u and D_f whose stencil nodes see different material values, and leaves every other row the standard one. For a tanh edge that region reaches about 19δ from each edge centre, where the tails round to the far-field value in double precision, plus the stencil’s reach. In each rebuilt row the $2p$ states of (9) for each field are integrated from $\xi = 0$ outward to each side with an explicit adaptive Runge–Kutta method of order eight, at relative tolerance 10^{-13} and absolute tolerance 10^{-15} , with segment boundaries at the edge centres and at $\pm 10\delta$ from them, so that the adaptive step never has to discover the edge in the middle of a long step across flat material. One march serves both fields and all p seeds, and a stencil that straddles both edges of a thin layer needs nothing further. The two $p \times p$ solves follow. In the reference implementation a march costs 5 to 15 ms, and the $n = 1600$ operator at $\delta = 0.01$ takes 2.3 s to build: seconds rather than milliseconds, fine for a study and not tuned. Two remarks on the range of δ . Since the seeds converge to the translated basis at first order in δ , an edge below about $10^{-5}h$ is better served by the interface-aware weights directly; and for δ of the order of the stencil or wider every row is rebuilt and the standard weights would have done as well (Section 3.4), so the rule of when to seed, settled in two dimensions in Section 6, applies here too.

4. ONE-DIMENSIONAL RESULTS

The experiment of this section is the one-dimensional test of the prediction that closes Section 3.3: standard FD4 loses its order through an edge it does not resolve and recovers it at $h \approx \delta$, while the seed stencils on the same grid are fourth order at every resolution with a constant that does not depend on δ .

Test problem. The medium is that of Section 2.1, the layer $[0, \frac{1}{2})$ with c from 1 to 2 at $\rho = 1$, so that K quadruples and the impedance doubles across each edge, with edges of width $\delta \in \{0, 0.0025, 0.01, 0.04\}$. The initial condition is a right-going Gaussian stress pulse, $f = \exp(-60(x+0.6)^2)$ and $u = -f/\rho c$, started in the background far enough from the layer that the tails of the reference below are clean. Its e^{-1} width is $2/\sqrt{60} \approx 0.26$, thirteen spacings on the coarsest grid, so the pulse is resolved on every grid and the edges are the only thing under test. The grids have $n = 100, 200, 400, 800$ and 1600 nodes, $h = 0.02$ down to 0.00125; for $\delta = 0.0025$ the ratio h/δ runs from 8 to $\frac{1}{2}$, for $\delta = 0.01$ from 2 to $\frac{1}{8}$, and for $\delta = 0.04$ every grid resolves the edge, $h/\delta \leq \frac{1}{2}$. Both schemes step with RK4 at $\Delta t = 0.4 h/c_{\max}$ to $t = 1$, by which time the pulse has crossed the layer: the transmitted pulse is in the background beyond the layer, the reflection from the lower edge is

TABLE 1. Relative ℓ^2 error in f at $t = 1$ against the reference, and the convergence rates between successive grids, for standard FD4 and the seed stencils (the interface-aware stencils at $\delta = 0$) on the same grids.

δ	scheme	$n = 100$	200	400	800	1600	rates
0 (jump)	standard FD4	7.1×10^{-2}	3.2×10^{-2}	1.58×10^{-2}	7.9×10^{-3}	3.9×10^{-3}	1.2, 1.0, 1.0, 1.0
0 (jump)	interface-aware	3.9×10^{-3}	2.5×10^{-4}	1.58×10^{-5}	9.9×10^{-7}	6.2×10^{-8}	4.0, 4.0, 4.0, 4.0
0.0025	standard FD4	7.2×10^{-2}	2.9×10^{-2}	6.3×10^{-3}	3.0×10^{-5}	7.4×10^{-6}	1.3, 2.2, 7.7, 2.0
0.0025	seeds	3.9×10^{-3}	2.5×10^{-4}	1.57×10^{-5}	9.9×10^{-7}	6.2×10^{-8}	4.0, 4.0, 4.0, 4.0
0.01	standard FD4	3.0×10^{-2}	5.6×10^{-4}	3.5×10^{-5}	1.05×10^{-6}	6.5×10^{-8}	5.7, 4.0, 5.0, 4.0
0.01	seeds	3.9×10^{-3}	2.5×10^{-4}	1.55×10^{-5}	9.7×10^{-7}	6.1×10^{-8}	4.0, 4.0, 4.0, 4.0
0.04	standard FD4	3.5×10^{-3}	2.2×10^{-4}	1.41×10^{-5}	8.8×10^{-7}	5.5×10^{-8}	4.0, 4.0, 4.0, 4.0
0.04	seeds	3.5×10^{-3}	2.2×10^{-4}	1.40×10^{-5}	8.8×10^{-7}	5.5×10^{-8}	4.0, 4.0, 4.0, 4.0

on its way back, and the reflection from the upper edge is still inside the layer (Figure 3). The error is the relative ℓ^2 error in f over the whole grid at $t = 1$.

References. For the jump the reference is the exact solution, a ray sum over the reflections and transmissions at the two interfaces. For $\delta > 0$ the coefficients are smooth and periodic and the reference is a Fourier pseudo-spectral discretisation of (1): derivatives in Fourier space, the coefficients applied pointwise, the same RK4 loop. It shares nothing with the finite-difference stencils under test, and the semi-discretisation conserves the energy $\int (f^2/K + \rho u^2) dx$ exactly, the Fourier derivative being skew-symmetric. Its resolution is set by the edge: on the interval of period 2, $\tanh(x/\delta)$ has poles at $\pm i\pi\delta/2$, so its Fourier coefficients decay like $e^{-\pi^2\delta m/2}$ in the mode number m , and the grid is the power of two that carries every mode down to 10^{-14} : 8192 nodes for $\delta = 0.0025$, 2048 for 0.01, and the floor of 1024 for 0.04. With $\Delta t = 5 \times 10^{-5}$ the reference is accurate to about 10^{-10} , below every error in Table 1 by more than two orders of magnitude, and is interpolated to the test grids trigonometrically.

The knee. Table 1 and Figure 2 give the errors and the rates between successive grids. For the jump the picture is the known one [Mar16, MF17a]: standard FD4 is first order and the interface-aware stencils are fourth order, 3.9×10^{-3} down to 6.2×10^{-8} . Through the edge of width $\delta = 0.0025$ standard FD4 converges at rates 1.3 and 2.2 while $h > \delta$, then drops by a factor 200 between $n = 400$ and $n = 800$, from $h = 2\delta$ to $h = \delta$: the knee. Over the two doublings past $h = 2\delta$ its error falls by a factor 850, an average rate of 4.9, unevenly (7.7, then 2.0), because the nodes sample the $\tanh$ at positions that move with n and so does the jump the stencil effectively sees; this is also why the rates before the knee are not the clean 1.0 of the jump. For $\delta = 0.01$ the knee sits between $n = 100$ and 200, again at $h \approx \delta$, and the rates after it are 4.0, 5.0 and 4.0. For $\delta = 0.04$, which every grid resolves, there is no knee and standard FD4 is fourth order throughout.

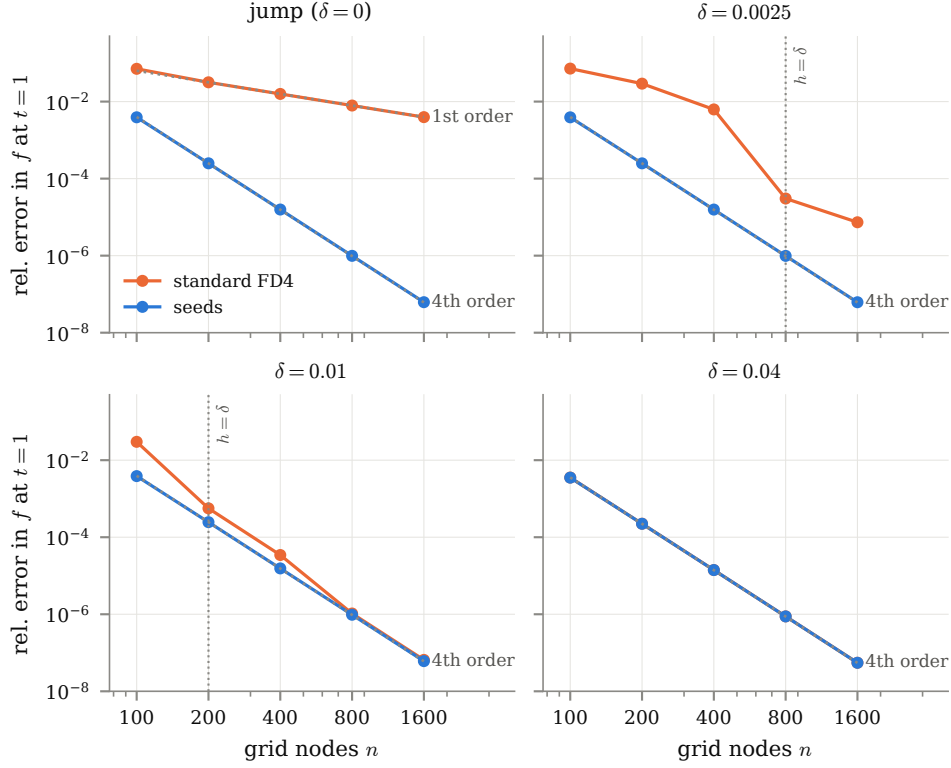


FIGURE 2. The errors of Table 1 against n for the four edge widths, with first- and fourth-order guides and, for $\delta > 0$, the grid at which $h = \delta$. Standard FD4 has its knee there; the seed stencils lie on one fourth-order line in every panel.

The seed stencils are fourth order at every n in every panel. Their errors through $\delta = 0.0025$ agree with the interface-aware errors of the jump case to three digits at every n , and through $\delta = 0.01$ to within 2%: the constant is the one of the jump construction, which is what Section 3.3 predicted, since the truncation error $O(h^4 \partial_x \partial_t^4 u)$ does not see the edge width. At $h = 8\delta$ the seeds are $18\times$ more accurate than standard FD4, at $h = 2\delta$ $400\times$, and at $h = \delta/2$, the end of the sweep for $\delta = 0.0025$, still $120\times$. Through the resolved edge $\delta = 0.04$ the two schemes agree to three digits at every n although their weights differ at first order in h/δ (Section 3.4): both are exact on a five-dimensional space that approximates the solution to fourth order, and on a resolved edge it does not matter which.

Truncation error. The same story is told without the time stepping by the local truncation error of the two differentiation matrices on the reference solution's profile at $t = 1$, here for the narrower pulse of the dissertation (sharpness 600), as the largest error in u_x over the rebuilt rows relative to $\max |u_x|$. Through $\delta = 0.0025$ the seed rows have errors 9.9×10^{-3} ,

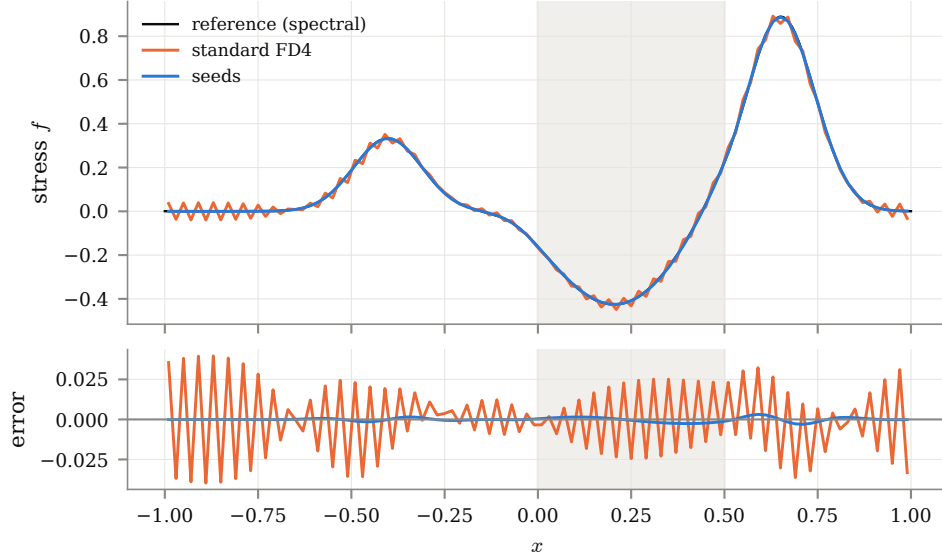


FIGURE 3. Stress f at $t = 1$ on 100 nodes through edges of width $\delta = h/8$ (the layer shaded), against the spectral reference, and the pointwise errors of the two schemes. The transmitted pulse has left the layer, the reflection from the lower edge is travelling back and the reflection from the upper edge is still inside; standard FD4 carries a grid-scale sawtooth everywhere, the seed stencils only the interior dispersion error.

1.2×10^{-3} , 9.1×10^{-5} , 5.8×10^{-6} and 3.6×10^{-7} at $h = 8\delta$, 4δ , 2δ , δ and $\delta/2$, successive ratios 8.5, 13, 16 and 16, fourth order from $h = 8\delta$ on; standard FD4 on the same rows falls only from 1.5×10^{-2} to 2.2×10^{-4} over the same five grids, and not monotonically. Through $\delta = 0.01$ the seed rows give 2.2×10^{-2} down to 2.3×10^{-7} on the same five grids: the same constant.

The coarse-grid picture. Figure 3 is the run at $n = 100$ through edges of width $\delta = 0.0025 = h/8$, so that both edges of the layer fall between nodes. Standard FD4 trails a grid-scale sawtooth of about 4% of the pulse height over the whole interval, ahead of the pulses as well as behind them: the highest modes of the FD4 stencil have negative group velocity, so the noise born at the edges outruns the physical waves. The seed stencils sit on the reference with a maximum error of 0.3% of the pulse height, which is the dispersion error of interior FD4 on the pulse itself.

Conditioning and stability. The 5×5 interpolation matrices of the rebuilt rows are the perturbed Vandermonde matrices of Section 3.1, with condition numbers between 20 and 60 for δ from 10^{-5} to 1; nothing in the sweep approaches the range of δ where the interface-aware weights should be used

directly. The semi-discrete operator of the seed scheme has no growing modes to speak of: the largest real part of its spectrum is 8×10^{-8} at $\delta = 0.005$ on 400 nodes and about 10^{-13} in the other cases examined, and RK4 at $\Delta t = 0.4 h/c_{\max}$ was stable in every run of the sweep. The runs cost seconds (Section 3.5). Nothing was tried at sharper contrasts, for layers thinner than the 19δ tails of an edge, or at orders above four; the construction does not change for any of them, and Section 7 lists them among the limitations.

5. SEEDS IN TWO DIMENSIONS

The chain of Section 3.1 is dimension-free: for any operator $\mathcal{L}$ the nested spaces $\{1\} \subset \ker \mathcal{L} \subset \{\mathcal{L}g = \text{const}\} \subset \ker \mathcal{L}^2 \subset \dots$ are what they were. What is not free is the choice of basis. In two dimensions $\ker \mathcal{L}$ is infinite-dimensional, and a space of “polynomial-like functions of degree $\leq q$ ” that reduces to the polynomials for constant coefficients needs a rule for which $\mathcal{L}$ -harmonic functions play the role of $1, x, y, xy, x^2 - y^2, \dots$: each harmonic polynomial needs a continuation through the edge that looks like it away from the edge. For a straight edge that rule is an ansatz, and it turns the problem back into ordinary differential equations in the normal coordinate. This section constructs the seeds of the elastic system for a straight edge, assembles them into stencils, and says what a curved edge does to the construction.

5.1. Design: a straight edge. A stencil that sees an edge works in the local frame of Section 2.3: origin at the nearest point of the edge centre, x along the tangent and y along the normal, the rotated fields obeying (5) unchanged. In this section (x, y) are those local coordinates and (u, v) and σ the rotated fields, the primes of Section 2.3 dropped. The material depends on y alone: exactly, for a flat edge, and for a curved one in the approximation of Section 5.4. Eliminating the stresses from (5), with $K = \lambda + 2\mu$,

$$(12) \quad \begin{aligned} \rho u_{tt} &= \partial_x [K u_x + \lambda v_y] + \partial_y [\mu(u_y + v_x)], \\ \rho v_{tt} &= \partial_x [\mu(u_y + v_x)] + \partial_y [\lambda u_x + K v_y], \end{aligned}$$

and $\mathcal{L}(u, v)$ denotes the right-hand side divided by ρ . The three brackets are the stress rates of (5), and the two that carry ∂_y are the tractions on the edge, σ_{xy} and σ_{yy} , which are what stays continuous across a jump. That is the 2-D form of the observation that drove the 1-D construction: the flux $K\phi'$, not the slope, is the smooth quantity through the edge.

Proposition 5.1 (Reduction to the normal coordinate). *Let λ, μ, ρ depend on y only, and let $u = \sum_{j=0}^q a_j(y) x^j$ and $v = \sum_{j=0}^q b_j(y) x^j$. Write the x^j -coefficients of the three stress rates as*

$$(13) \quad \begin{aligned} F_j &= K(j+1) a_{j+1} + \lambda b'_j, & \tau_j &= \mu(a'_j + (j+1) b_{j+1}), \\ \nu_j &= \lambda(j+1) a_{j+1} + K b'_j, \end{aligned}$$

for σ_{xx} , σ_{xy} and σ_{yy} respectively, with $a_j = b_j = 0$ for $j > q$. Then $\rho\mathcal{L}(u, v)$ is a polynomial in x of degree $\leq q$ whose x^j -coefficients are

$$(14) \quad \rho\mathcal{L}_{1,j} = (j+1)F_{j+1} + \tau'_j, \quad \rho\mathcal{L}_{2,j} = (j+1)\tau_{j+1} + \nu'_j.$$

In particular the coefficients at level j involve (a, b) at levels j , $j+1$ and $j+2$ only, so an equation $\mathcal{L}(u, v) = (\text{given})$ is a triangular system, from the top x -degree down, of two-component second-order ODEs in y for the pairs (a_j, b_j) .

Proof. Each bracket in (12) is a polynomial in x with coefficient functions of y , as displayed in (13); ∂_x shifts the index and multiplies by it, ∂_y differentiates the coefficient, and (14) follows by reading off. Expanding F_{j+1} gives $\rho\mathcal{L}_{1,j} = (j+1)(j+2)Ka_{j+2} + (j+1)\lambda b'_{j+1} + \tau'_j$, which reaches level $j+2$; τ'_j and ν'_j involve a''_j , b''_j and b'_{j+1} , a'_{j+1} , and nothing lower. $\square$

First-order form. As in one dimension, the chain is integrated as a first-order system that never differentiates the coefficients, and the tractions are the flux variables. Solving (13) for the slopes and (14) for the derivatives of the tractions, the state at level j is $(a_j, b_j, \tau_j, \nu_j)$ with

$$(15) \quad \begin{aligned} a'_j &= \tau_j/\mu - (j+1)b_{j+1}, \\ b'_j &= (\nu_j - \lambda(j+1)a_{j+1})/K, \\ \tau'_j &= \rho R_{1,j} - K(j+2)(j+1)a_{j+2} - \lambda(j+1)b'_{j+1}, \\ \nu'_j &= \rho R_{2,j} - (j+1)\tau_{j+1}, \end{aligned}$$

where b'_{j+1} in the third line is the second line at level $j+1$ and (R_1, R_2) is the right-hand side the chain prescribes, next. The coefficients λ , μ , ρ are only ever evaluated. The pair (τ_j, ν_j) is the 2-D twin of $\psi_k = K\phi'_k$ in (9).

5.2. The elastic seeds. With constant coefficients $\mathcal{L}$ maps a pair of monomials to pairs of lower degree,

$$(16) \quad \begin{aligned} \mathcal{L}(x^a y^b, 0) &= \frac{1}{\rho} \begin{pmatrix} Ka(a-1)x^{a-2}y^b + \mu b(b-1)x^a y^{b-2} \\ (\lambda + \mu)abx^{a-1}y^{b-1} \end{pmatrix}, \\ \mathcal{L}(0, x^a y^b) &= \frac{1}{\rho} \begin{pmatrix} (\lambda + \mu)abx^{a-1}y^{b-1} \\ \mu a(a-1)x^{a-2}y^b + Kb(b-1)x^a y^{b-2} \end{pmatrix}, \end{aligned}$$

that is, a matrix C over the $2m$ monomial pairs e (m monomials to degree q , the u -block then the v -block), $\mathcal{L}e = \sum_{e'} C[e', e] e'$, which lowers the degree by two and is therefore strictly triangular in degree. It is the (u, v) block of the square of the block operator of (5) on polynomial coefficients, the same object the continuity matrices of Section 2.3 are built from.

Definition 5.2 (Elastic seeds of a straight edge). Let the evaluation node be at (x_e, y_e) in the local frame and let C be the matrix (16) for the material

at y_e . The *seeds* S_e , one per monomial pair e , are the solutions of

$$(17) \quad \mathcal{L}S_e = \sum_{e'} C[e', e] S_{e'}$$

of the form of Proposition 5.1 in $(x - x_e, y - y_e)$, with the monomial's jet as initial data at $y = y_e$: for $e = (x^\alpha y^\beta, 0)$, $a_\alpha(y_e) = 1$ if $\beta = 0$, $a'_\alpha(y_e) = 1$ if $\beta = 1$, and every other a_j , b_j and slope zero at y_e , with the mirror image for $e = (0, x^\alpha y^\beta)$; the tractions at y_e follow from the jet and the material there through (13).

This is Definition 3.1 read in two dimensions: $L_u \phi_k = k(k-1) c_e^2 \phi_{k-2}$ is (17) with C the 1-D chain matrix, and the anchor conditions are the jet. The seeds of the lower monomials stand on the right, not the monomials, which is what makes the system consistent: by the triangularity of C a seed only ever needs seeds two degrees down, and all $2m$ of them march together as one linear system. With constant coefficients the right-hand side of (17) is $\mathcal{L}$ of the monomial itself, by induction on the degree, and the linear ODE with that data has one solution, the monomial: the seeds are the monomials, to 2.5×10^{-14} in the implementation's test. For the seed of $(x^\alpha y^\beta, \cdot)$ only the levels $j \leq \alpha$ are ever non-zero, since nothing above them is driven, so a seed keeps the x -degree of its monomial and the medium enters through the y -structure alone. Two seeds are in closed form through any edge: $(y, 0)$ becomes $(\int_{y_e}^y \mu_e / \mu \, d\eta, 0)$ and $(0, x)$ becomes $(\int_{y_e}^y (\mu_e / \mu - 1) \, d\eta, x)$, both with the constant shear traction μ_e and no other stress. That is the twin of $\phi_1 = K_e \int d\xi / K$: the traction is what stays smooth, not the slope.

The system for degree three. The stencils across an edge are exact on velocities to degree $p = 3$, and the stress basis is generated from velocities one degree higher, as in the jump case, so the seeds are marched for the monomials to degree $q = 4$: 15 monomials per component, 30 seeds, x -degrees 0 to 4, and five levels of $(a_j, b_j, \tau_j, \nu_j)$ per seed, a linear system of 600 states. It is integrated with the same explicit eighth-order Runge–Kutta method and tolerances as in one dimension, from the anchor outward to each side, restarted at every node's normal coordinate so that the node values are integrated rather than interpolated, and at the edge centres and their $\pm 10\delta$ flanks. The flanks matter here as they did in one dimension: a single solve without them differs from the segmented march by 4×10^{-10} at $\delta = 10^{-4}$, where the adaptive step has to discover the edge.

Stress seeds and rigid motions. The stress seed of a velocity seed has the components $\sigma_{xx} = \sum_j F_j x^j$, $\sigma_{xy} = \sum_j \tau_j x^j$ and $\sigma_{yy} = \sum_j \nu_j x^j$, read off the marched state and the flux variables, with b'_j in F_j taken from the second line of (15), so nothing is differentiated numerically. As in the jump construction (Section 2.3), the constants in u and v have no stress and the shear pair $(y, 0)$, $(0, x)$ share theirs, through any edge and not only for constant coefficients, so the same three columns are dropped: 27 stress seeds from the 30 velocity

seeds, and the 20 velocity seeds of degree ≤ 3 for the velocity block, the dissertation's counts.

Anchor, scaling and jets. The seeds are anchored at the evaluation node in both coordinates and scaled by $r_{\max}$, the distance to the farthest node of the stencil, so that $S_e \approx X^a Y^b$ in $(X, Y) = (x - x_e, y - y_e)/r_{\max}$ and the material is sampled at $y_e + r_{\max} Y$. The jump basis of Section 2.3 keeps its origin at the interface point. Moving the origin in x is a change of basis within the same span and changes no weight, and the two conventions are reconciled that way, so the jump path is untouched. Where the *normal* coordinate is anchored is not a change of basis, because that is where the jet is imposed and the material is frozen in C ; there the seeds follow the lesson of one dimension, that anchoring at the evaluation point is what keeps the interpolation block conditioned. At the anchor the velocity seeds have the monomials' first derivatives (only X and Y have one), and the derivatives of the stress seeds come from the state and the right-hand side of (15) at $Y = 0$, so the elastic rates of every basis function at the evaluation node are known exactly. One of the implementation's tests is the identity this implies: the velocity rate of a stress seed at the anchor, $(\partial_X \sigma_{xx} + \partial_Y \sigma_{xy})/\rho_e$, is the constant term of the seed's column of C .

Checks. Three properties were checked on a 19-node stencil of a 400-node set ($h = 0.05$) on the row nearest the lower interface, the anchor $h/2$ into the band or $h/2$ into the background. *Jump limit:* as $\delta \rightarrow 0$ the state passes through the edge unchanged, which is velocity and traction continuity, and on the far side the chain holds with the anchor's frozen coefficients, which is what the continuity matrices encode; so the seeds should tend to the translated basis of the jump case re-expanded about the anchor, and they do, at first order in δ : the largest difference relative to the largest basis value is 1.0 to 2.3% at $\delta = 10^{-3}$ and falls tenfold per decade to 1.0 to 2.4×10^{-4} at $\delta = 10^{-5}$, for the velocity and the stress basis and with the anchor on either side. *Residual:* every seed, evaluated as a function of (x, y) on a 25×401 tensor grid through a $\delta = 0.01$ edge and pushed through the elastic operator by eighth-order finite differences in both variables, satisfies (17) with relative residual 2×10^{-12} ; the test uses none of the module's coefficient bookkeeping, and the residual rises to 0.23 if the factor $(j+2)(j+1)$ in (15) is wrong and to 0.06 if the $\lambda(j+1)b'_{j+1}$ term is dropped. *Conditioning:* the 2-norm condition numbers of the velocity block (38×20 : two components at 19 nodes against 20 seeds) and the stress block (57×27) are 17 to 34 and 24 to 30 for δ from h down to $h/5000$, against 34 and 30 for the polynomial jump basis re-expanded about the same anchor, and they tend to those values as $\delta \rightarrow 0$. No Chebyshev-system argument is available in two dimensions, where interpolation on scattered nodes can be singular even for polynomials. The saddle-point solve is nonsingular when the Gaussian block is positive definite, which it is, and the augmentation has full column rank; for the

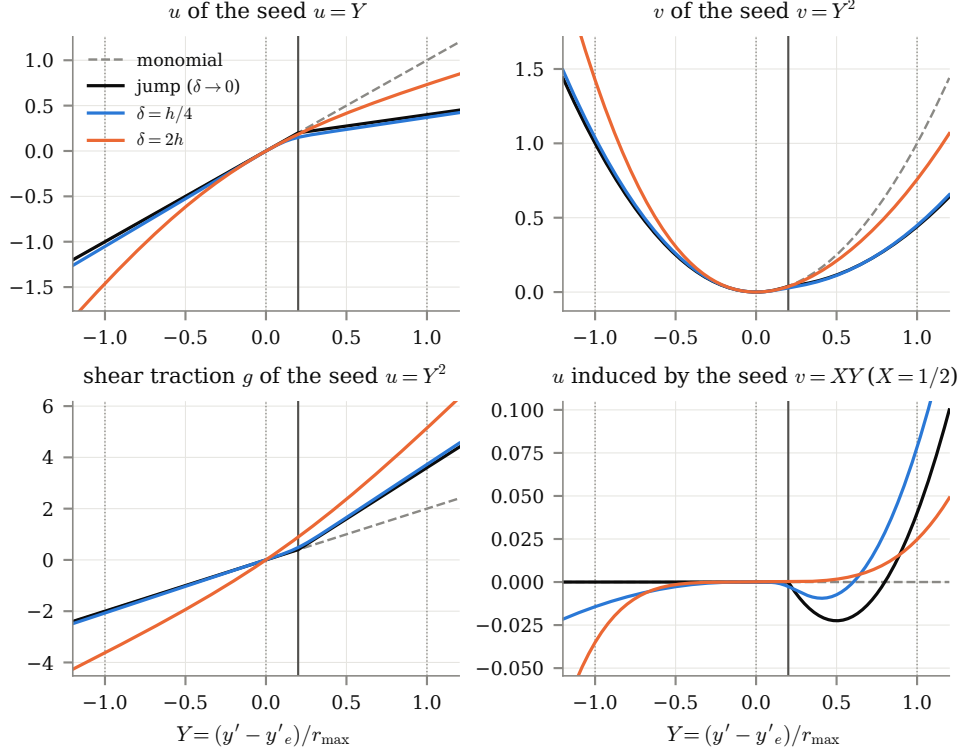


FIGURE 4. Cross-sections along the normal through the anchor of four elastic seeds of a 19-node stencil whose evaluation node lies $h/2$ below the lower edge of the band of Section 2.2 (λ, μ from 1 to 4, ρ from 1 to 2), in the stencil coordinate $Y = (y - y_e)/r_{\max}$ with $r_{\max} = 2.5h$, for the jump limit (computed at $\delta = 10^{-6}$), $\delta = h/4$ and $\delta = 2h$, against the monomials of the anchor material (dashed): the u of the seed $u = Y$; the v of the seed $v = Y^2$; the shear traction of the seed $u = Y^2$ (labelled g , the implementation's name for the rate of σ_{xy}); and the u induced by the seed $v = XY$ at $X = \frac{1}{2}$, which is zero for the monomial and is the tangential coupling that the ablation of Section 6 removes. The vertical line is the edge centre and the dotted lines the stencil's reach $|Y| \leq 1$. As in Figure 1, the $\delta = h/4$ seeds are close to the jump limit and the $\delta = 2h$ seeds bend over the whole stencil.

seeds the measured conditioning is the evidence for the latter, and it is all we have.

5.3. Stencil assembly. Which rows. The seed scheme rebuilds the rows of every node whose 19 nearest nodes see different values of λ, μ or ρ , the 1-D rule of Section 3.5 on the stencil's footprint: with exact inequality that

reaches about 19δ from each edge centre plus the stencil's radius, and a relative tolerance on the spread trims the tails (the trim of Section 6). The nearest interface, by vertical offset, sets the frame of a rebuilt row; the material profile along its normal carries both edges of the band, so a stencil that sees both is marched through both, and the triple stencils of the jump case are not needed.

Elastic rows. The rebuilt elastic rows are the coupled saddle-point solves of Section 2.3 with the seeds in place of the piecewise polynomials: the Gaussian block on the 19 nodes is unchanged, the augmentation is the 38×20 block of velocity-seed values and the 57×27 block of stress-seed values, the right-hand sides on the seed columns are the elastic rates of (5) at the evaluation node, from the jets, and the material in those rates is the anchor material, which is the node's own. The weights are rotated back to the laboratory frame as before. Every stencil's seeds are its own, even on a flat edge, because the jet and the frozen C live at its anchor.

Hyperviscosity rows. The Δ^3 rows of the same nodes are built on the naive footprint of 30 nodes, with the same seeds annihilated: the saddle-point solve imposes $\Delta^3 S_e = 0$ on every seed column, as it imposes $\Delta^3 p = 0$ on the polynomial augmentation of a standard stencil. That is a second march per stencil, over the larger radius. Two things decided the footprint, both measured in Section 6. A 19-node stencil carrying 20 coupled constraints has 18 degrees of freedom left for the Gaussian part of a sixth-order operator, and what comes out is barely a Δ^3 : on the 900-node set the extreme eigenvalue of its Δ^3 block is -82 against -151 for the standard rows, half the damping, and the operator built that way is unstable at the standard γ once the seeded region, 19δ plus the stencil radius, has grown to take in every row, from about $\delta = h/2$ on that node set; and the alternative of 30-node, degree-four stencils for the elastic rows themselves is stable through a resolved edge but violently unstable through a sharp one, as a degree-four basis across a jump is, which is presumably why the original work settled on 19 nodes and degree three. Annihilating the seed space is a choice, not a derivation: the seeds' true sixth derivatives at the anchor are not zero in a varying medium, and using them as right-hand sides is the alternative Section 7 records as future work.

Checks. Without contrast the seed weights equal the degree-three RBF-FD weights and the no-contrast weights of the jump path to 10^{-10} ; as $\delta \rightarrow 0$ they tend to the interface-aware weights at first order in δ on a real stencil, all four blocks, whichever side carries the standard monomials; every block is exact on the seed space through a $\delta = h/4$ edge and the Δ^3 blocks annihilate it; and the rebuilt rows of the assembled operator are exactly the stencils that see the edge, with no other entry moved. The jump path of Section 2.3 is unchanged bit for bit.

An ablation. To measure what the tangential structure of the seeds contributes, Section 6 uses a variant in which only the seeds of the pure normal

monomials y^b are marched through the edge and every $x^a y^b$ with $a \geq 1$ is the plain monomial with the anchor material's stress. The plain columns come from a second chain on the material frozen at the anchor, since a frozen column must see frozen lower seeds on its right-hand side or $x^2 y^2$ would still be driven by the marched y^2 . The span has the same dimension and the same jets.

Cost. A 19-node march costs 21 to 25 ms and a seed row, with its 30-node march, 53 ms, in the reference implementation. On 10,000 nodes the operator through $\delta = h/8$ has 1965 seed rows and builds in 105 s; through $\delta = 0.01 = h$ it has 6719 and builds in 331 s. The marches run on a process pool and the assembled operators are cached, which is what the sweeps of Section 6 needed. Two optimisations were not built: the 19-node stencil is a prefix of the 30-node one, so one march with the seeds rescaled to the smaller $r_{\max}$ would serve both rows; and on a flat edge every stencil of one fixed row shares y_e and the profile, so one march with dense output could serve a whole row.

5.4. Curved edges. For a curved edge the material no longer depends on the normal coordinate of one frame, and three routes were open at the design stage: (a) march the straight-edge seeds along the true normal through the stencil's foot point and treat the curvature as a perturbation, $\kappa h \ll 1$ over a stencil; (b) solve (17) as a small boundary-value problem on a patch around each stencil, on a fine local grid with the harmonic polynomial as boundary data, the oversampling construction of multiscale finite elements [HW97] and the honest "one PDE per stencil"; or (c) compute harmonic coordinates once globally [OZ07] and take polynomials in them as the degree-one seeds. Route (a) is what was built, at its zeroth order.

The medium. With a curved interface the blend of (6) is taken in the true signed normal distance: the foot point comes from the same fixed-point iteration that places the stencil frames, and the distance is the displacement's component along the normal there, so the foot point's error enters squared and the distance is exact to rounding wherever the iteration converges at all. The two edges of the band are read on one periodic image of it, the image whose centre line is nearest; reading each edge on its own nearest image pairs the lower edge of one band with the upper edge of the next around $y = 0.75$, and was how the first version failed its own test.

The seeds. The profile of a rebuilt row carries the tangent angle θ at its foot point (x_0, y_0) and samples the medium along the true normal, $(x_0 - y \sin \theta, y_0 + y \cos \theta)$, with the march's stops at the crossings of that line with each interface and its images, by Newton's method from the vertical estimate. Along that line the blend is exactly the medium's tanh profile in y for the interface the line is normal to (to 10^{-13}), so the seeds of a curved-edge stencil are the *flat* seeds of the same local coordinates (to 10^{-12} through a $\delta = 0.005$ edge on a stencil tilted by 0.08 radians): the ansatz extends the normal profile unchanged along the tangent. That is the locally

flat approximation the jump stencils of Section 2.3 make, zeroth order in the curvature, and nothing else in the construction changes: the frames, the anchor at the evaluation node, the $r_{\max}$ scaling, the 19-node elastic rows and the 30-node Δ^3 rows are those of Section 5.3. A curved march costs $3.6\times$ a flat one, 131 against 37 ms per 19-node stencil, because every material sample along the normal is a blend evaluation with two foot-point iterations and the march takes a few thousand of them.

What route (a) gets wrong. In the tangent frame a node at tangential offset x from the foot point lies at true normal distance $y - \kappa x^2/2 + O(\kappa^2)$ from the curve, while its seed value assumes y . Over a stencil radius $r \approx 2.5h$ that is $\kappa r^2/2$: for the sine interfaces of amplitude 0.02, whose curvature reaches $\kappa = 0.02(2\pi)^2 = 0.79$, it is 1.0×10^{-3} at 2500 nodes and 1.3×10^{-4} at 19,600, to be compared with δ : a tenth of a $\delta = 0.01$ edge at the coarse end and 5% of a $\delta = 0.0025$ edge at the fine end. So the geometric error of route (a) is largest where the seeds matter most, through the sharpest edge, and it shrinks with n only as h^2 . Section 6 measures what it costs on the node sets we can afford, and finds route (a) at the seed operator's own floor on all of them, so neither (b) nor (c) was built. If a sharper edge or a tighter curve ever asks for more, the cheaper first step is not the boundary-value problem but the curvilinear evaluation of the same marched seeds: evaluate each seed at the node's true arclength and normal distance instead of its tangent coordinates, and scale the tangential jets at the anchor by $1/(1 - \kappa y_e)$. That makes the material exact at every node at no march cost and leaves only the operator's curvature terms as the error; Section 7 lists it first among the things to do next.

6. TWO-DIMENSIONAL RESULTS

6.1. Test problems and references. *[Stub, #59: 1-D spectral at normal incidence; Fourier-in- x for oblique; product-grid for the curved $\delta > 0$; the 122,500-node jump-aware run for the curved $\delta = 0$.]*

6.2. Naive baseline. *[Stub, #59: the resolution floor hides the edge error in v ; the spurious u separates unresolved from resolved edges $10\times$ more sharply.]*

6.3. Stability. *[Stub, #59: spectra at $n = 900$ and 2500, standard γ , every δ , flat and curved.]*

6.4. The flat sweep and the rule. *[Stub, #59: seeds fourth order through $\delta = 0.0025$ at every n , $20\times$ below naive in v and $67\times$ in spurious u at 19,600 nodes; crossover at $h \approx \delta$; a resolved edge $1.4\text{--}2\times$ worse seeded; the rule: seed when $\delta \leq h$.]*

6.5. Oblique incidence. *[Stub, #59: seeds $1.4\text{--}1.9\times$ over naive at 26.6° , the ablation worse than naive, every scheme about 2.5 from the converted- S floor, reported as it is.]*

6.6. The curved edge. [Stub, #59: jump stencils at the resolution floor at every n ; seeds equal the flat seeds at their own floor, $2.2\text{--}7\times$ ($\delta = 0.005$) and $3\text{--}12\times$ ($\delta = 0.0025$) below naive; where route (a)'s geometry first shows; the fine-end rate and the trim; costs.]

7. DISCUSSION

7.1. Limitations. [Stub, #60: FD_4 / degree 3 only, one contrast, at most 19,600 nodes, the flat translation symmetry unexploited, one march per stencil, the port's exclusions.]

7.2. Future work. [Stub, #60: curvilinear evaluation of the marched seeds; seeded-width tuning; the seeds' true Δ^3 as hyperviscosity right-hand sides; Mühlbach's recurrence in place of the dense solve; higher order; other contrasts; thin bands; variable Lamé parameters; 3-D; the oblique train on the curved geometry.]

7.3. Conclusions. [Stub, #60: the rule and the one-construction claim, in #53's wording.]

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[Stub, #62: Bengt Fornberg, for the 2016-era work this construction extends (the dissertation and the interface treatment of [MF17a]); the AI-assistance statement crediting Claude (Fable 5.1) for the re-derivation, the code and the drafting.]

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